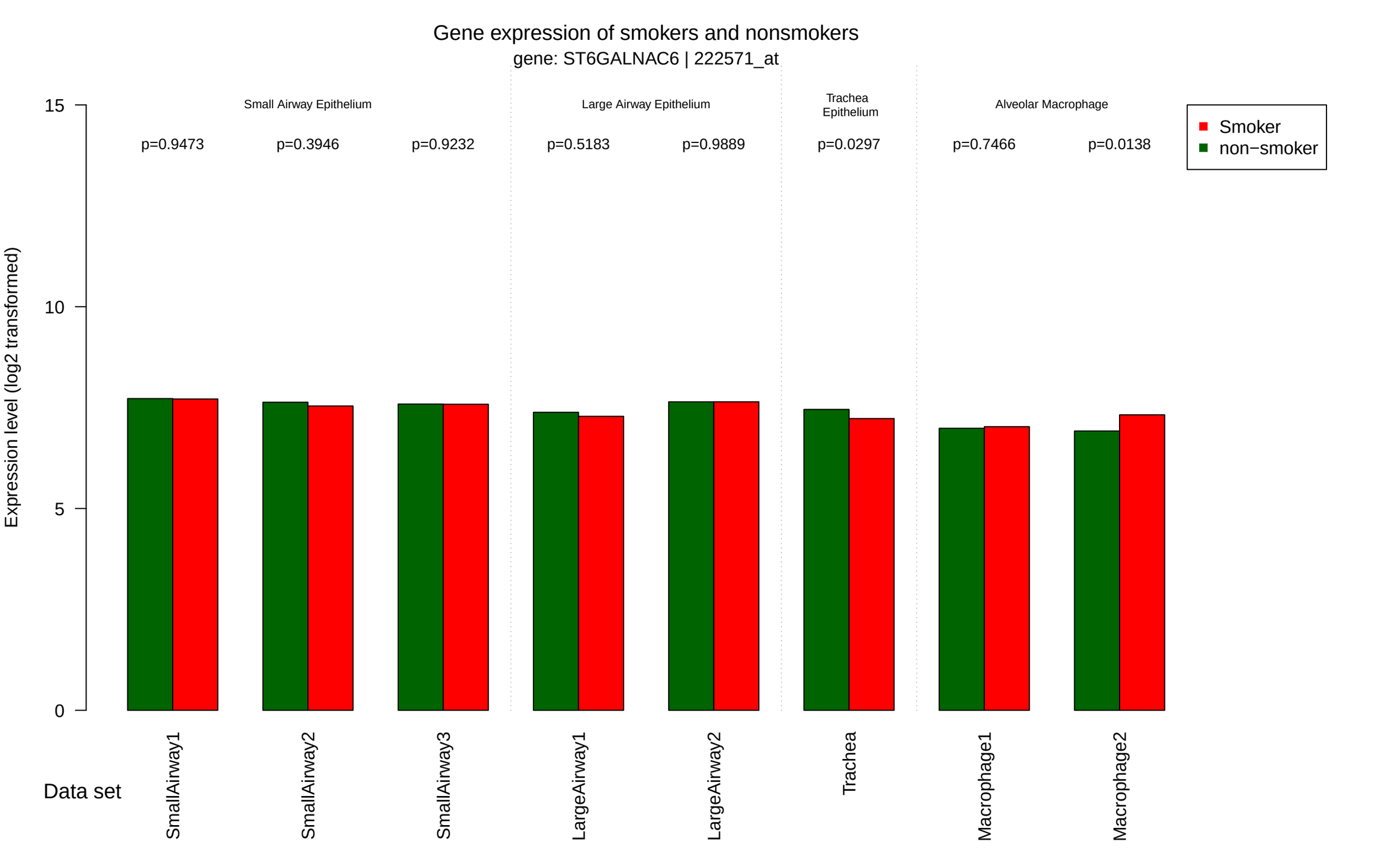
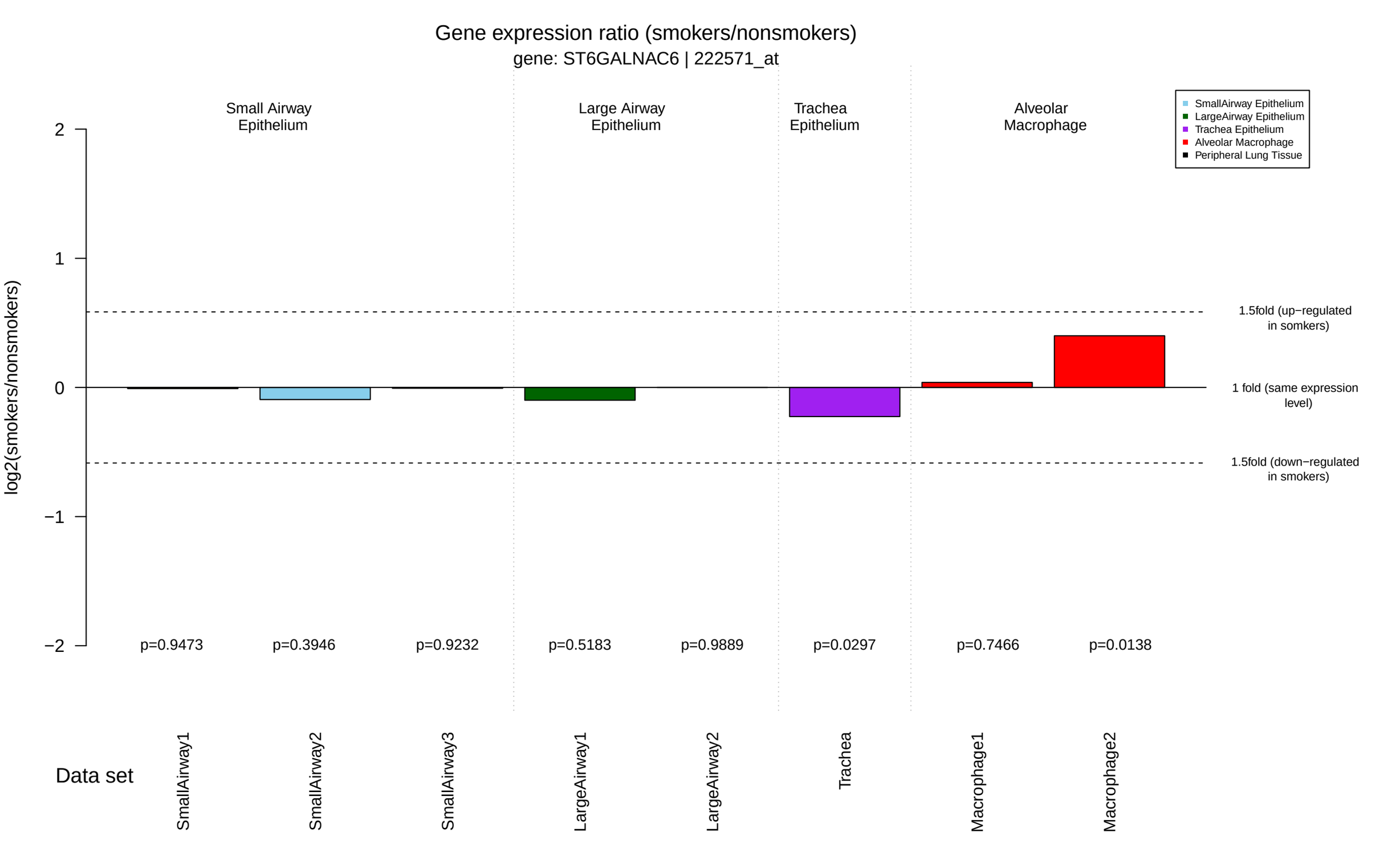
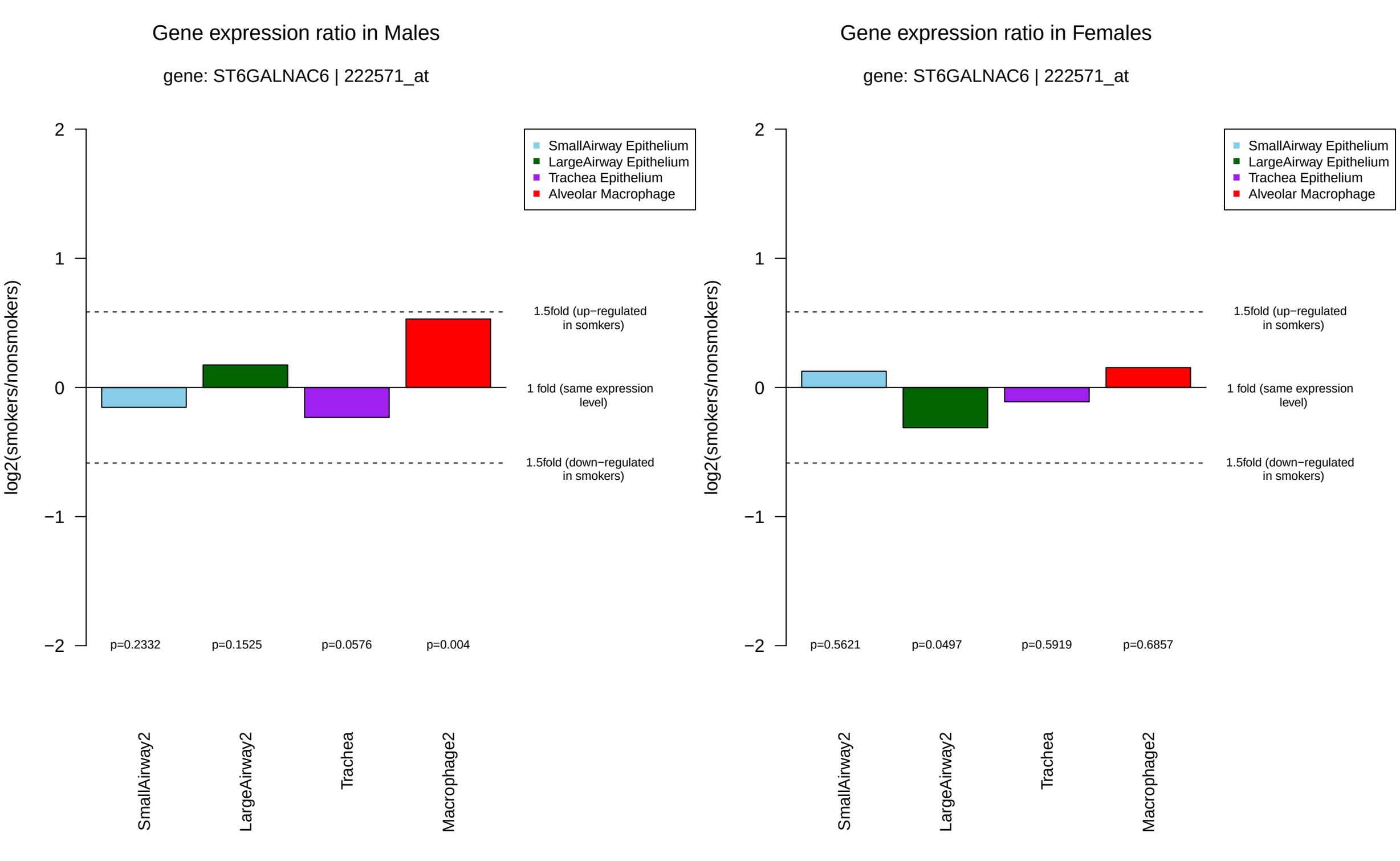
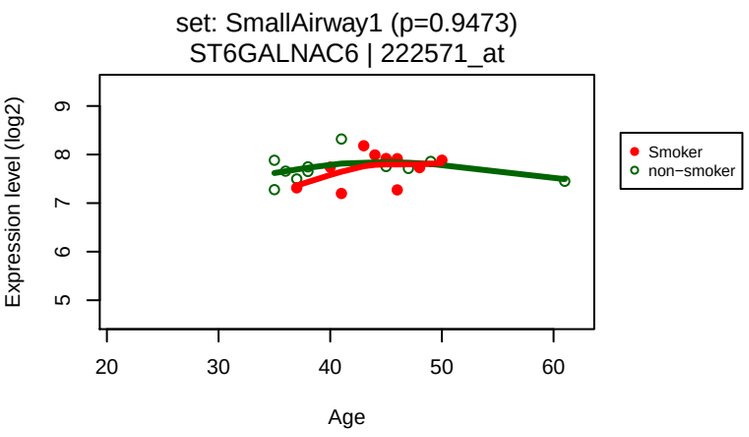
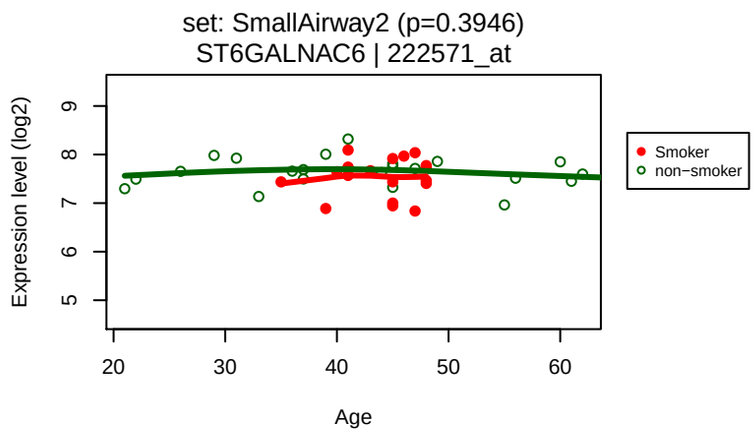
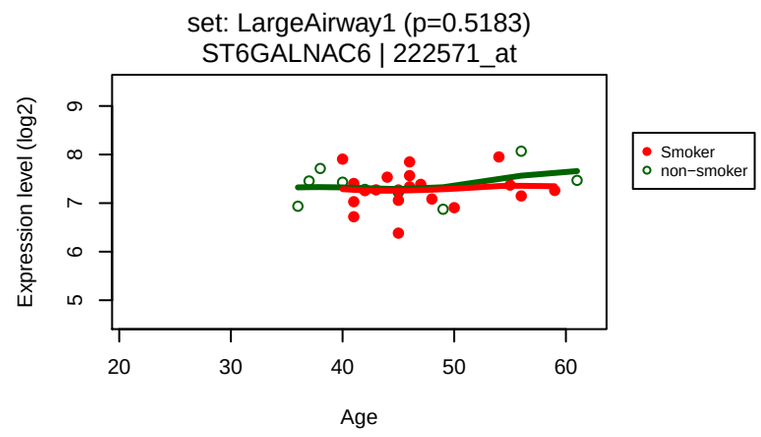

Gene expression of smokers and nonsmokers
gene: ST6GALNAC6 | 222571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9473
p=0.3946
p=0.9232
p=0.5183
p=0.9889
p=0.0297
p=0.7466
p=0.0138
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

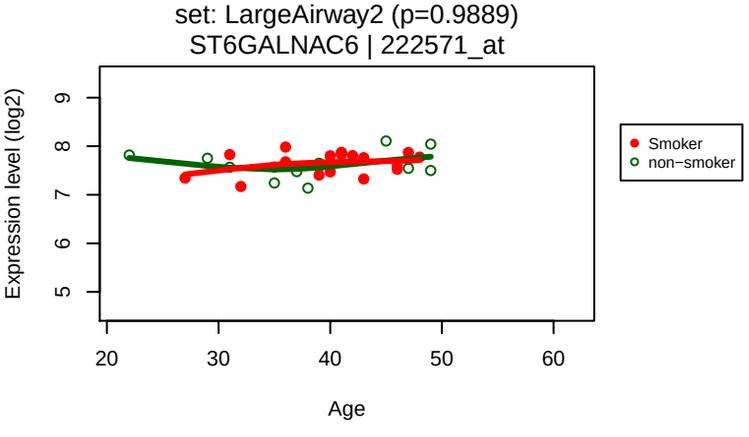

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC6 | 222571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9473
p=0.3946
p=0.9232
p=0.5183
p=0.9889
p=0.0297
p=0.7466
p=0.0138
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

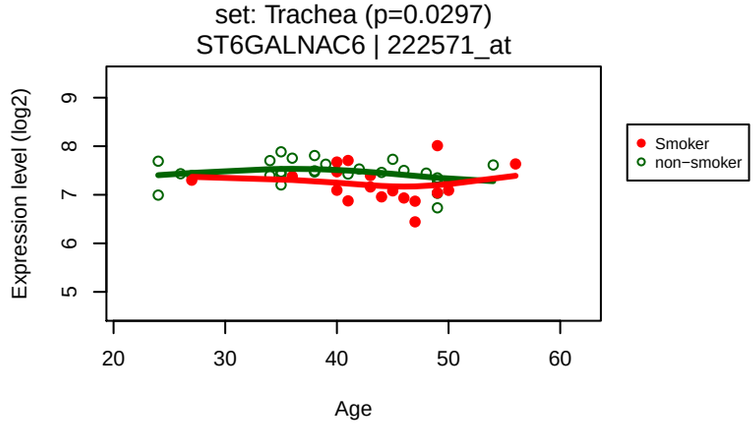

Gene expression ratio in Males
gene: ST6GALNAC6 | 222571_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2332
p=0.1525
p=0.0576
p=0.004
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC6 | 222571_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5621
p=0.0497
p=0.5919
p=0.6857
SmallAirway2
LargeAirway2
Trachea
Macrophage2

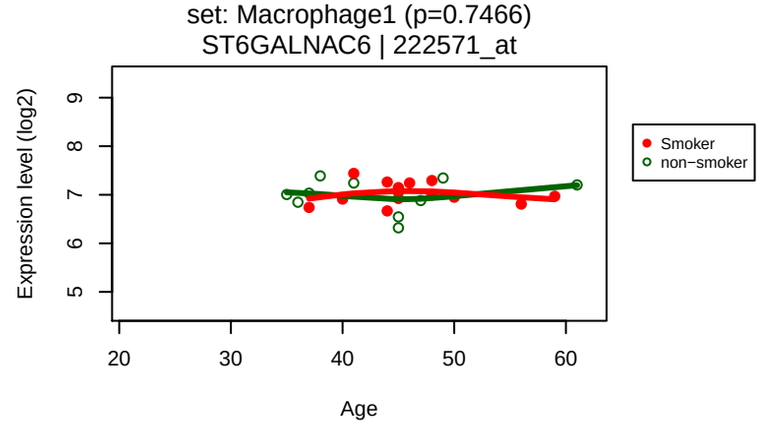

set: SmallAirway1 (p=0.9473)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker

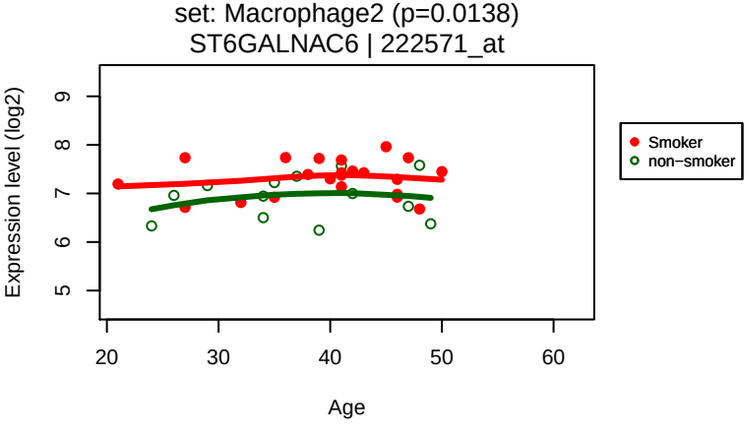

set: SmallAirway2 (p=0.3946)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker

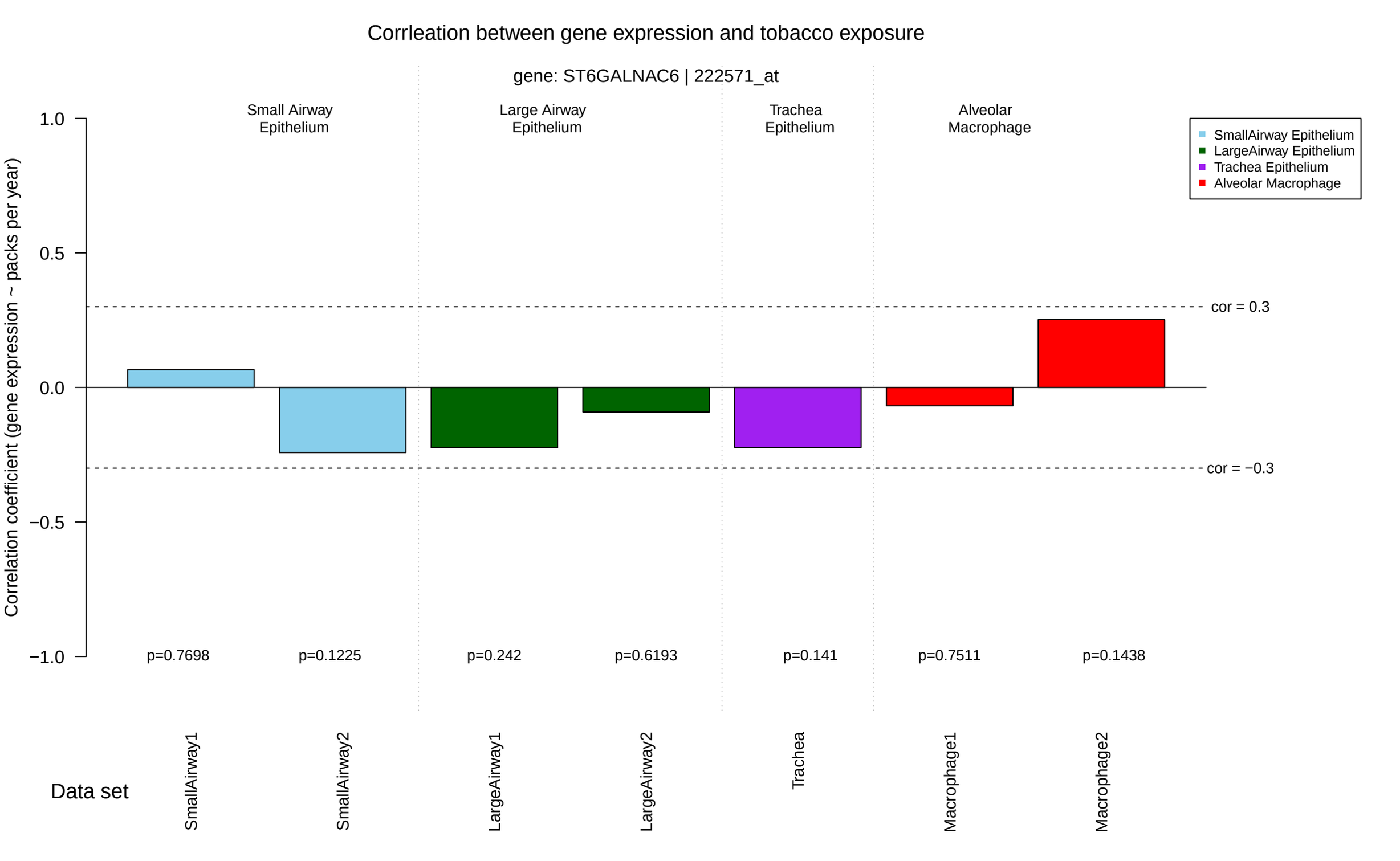

set: LargeAirway1 (p=0.5183)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9889)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0297)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7466)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0138)
ST6GALNAC6 | 222571_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC6 | 222571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7698
p=0.1225
p=0.242
p=0.6193
p=0.141
p=0.7511
p=0.1438
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2